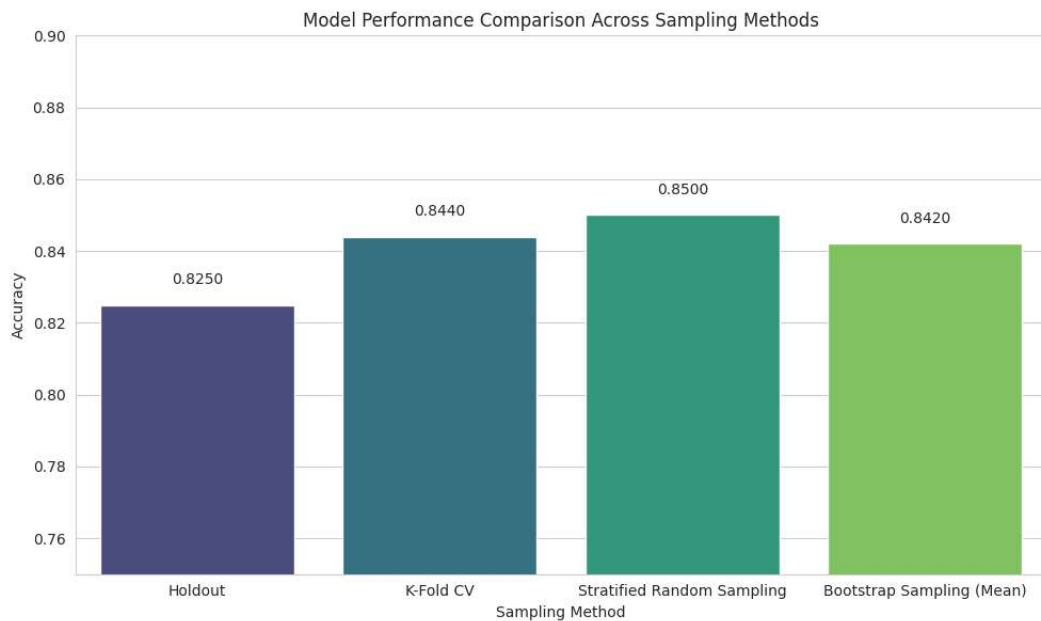
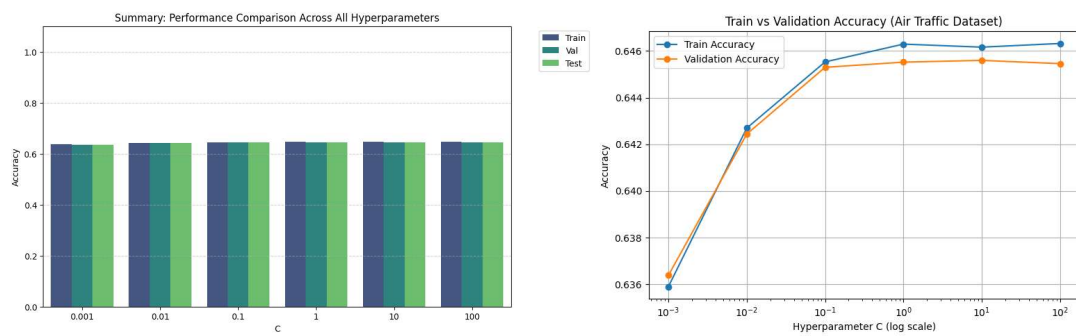




OUTPUT:

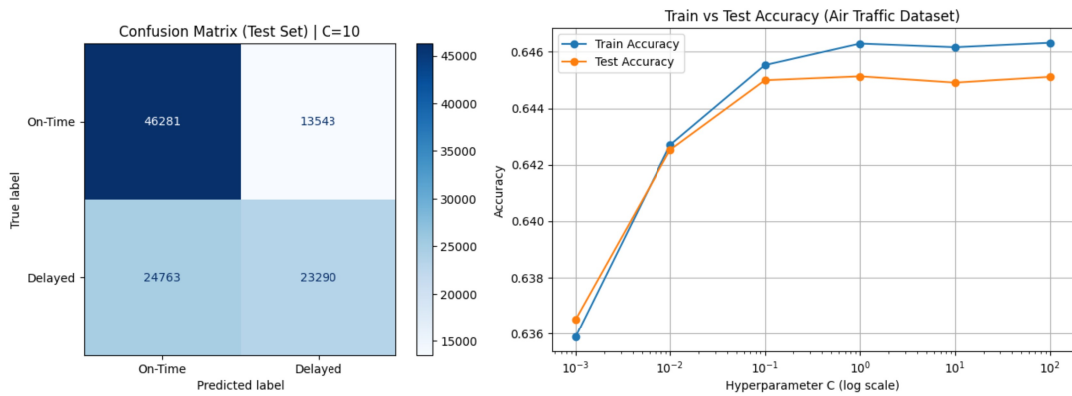


Best Hyperparameter Found:

Best C = 10

Best Validation Accuracy = 0.6455963736477656

C=10 | Train=0.6462 | Val=0.6456 | Test=0.6449



Dataset Shape: (539382, 8) Columns: ['Flight', 'Time', 'Length', 'Airline', 'AirportFrom', 'AirportTo', 'DayOfWeek', 'Class']

Unique Classes in y: [0 1] Split Sizes: Train: 323628 Validation: 107877 Test: 107877

Hyperparameter Tuning Results: Best Hyperparameter Found: Best C = 10 Best Validation Accuracy = 0.6455963736477656

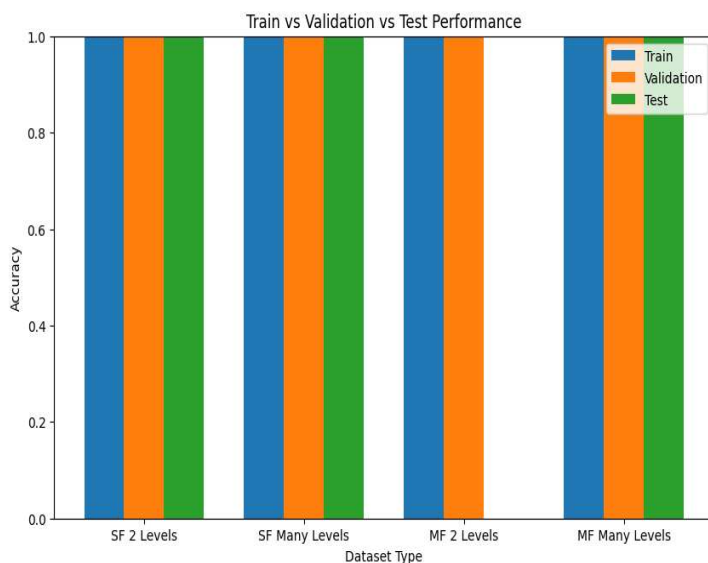
OUTPUT: Naive Bayes classifier successfully classified IoT sensor data.

```
*** --- 1. Single Feature, 2 Levels ---
Input: [{'Connectivity': 'Offline'}]
Output Prediction: ['Yes']

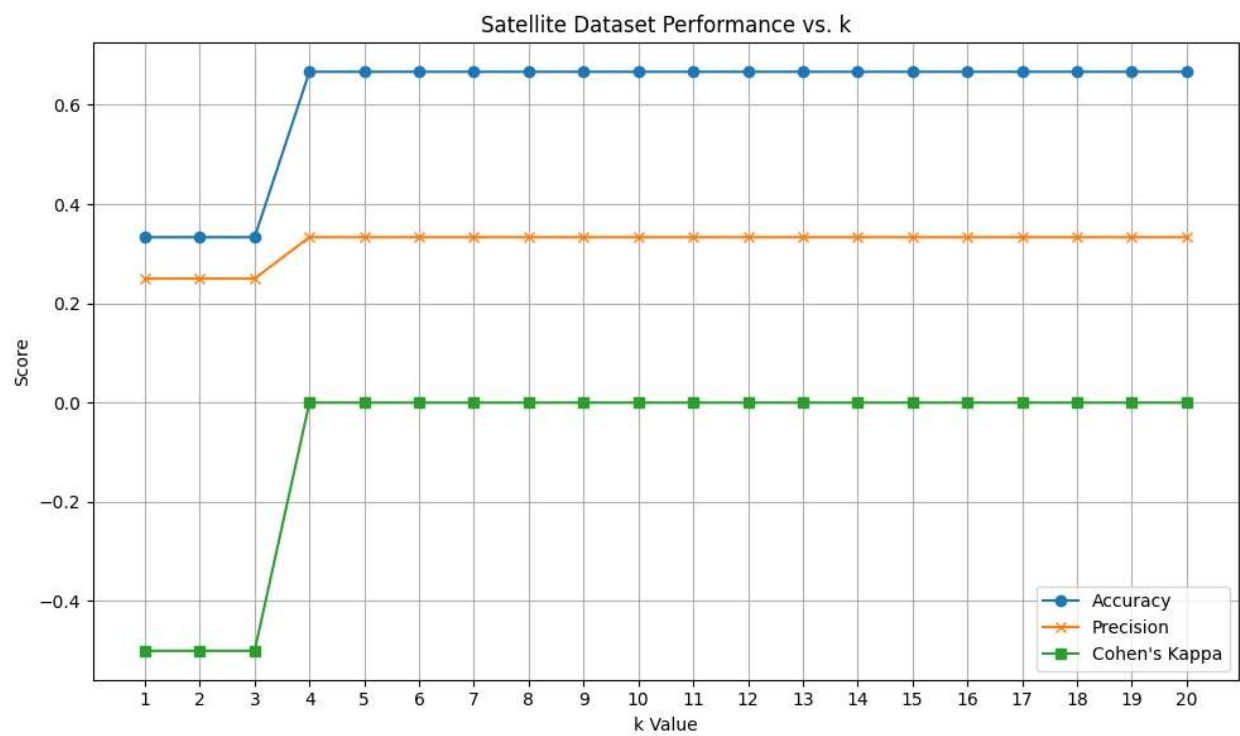
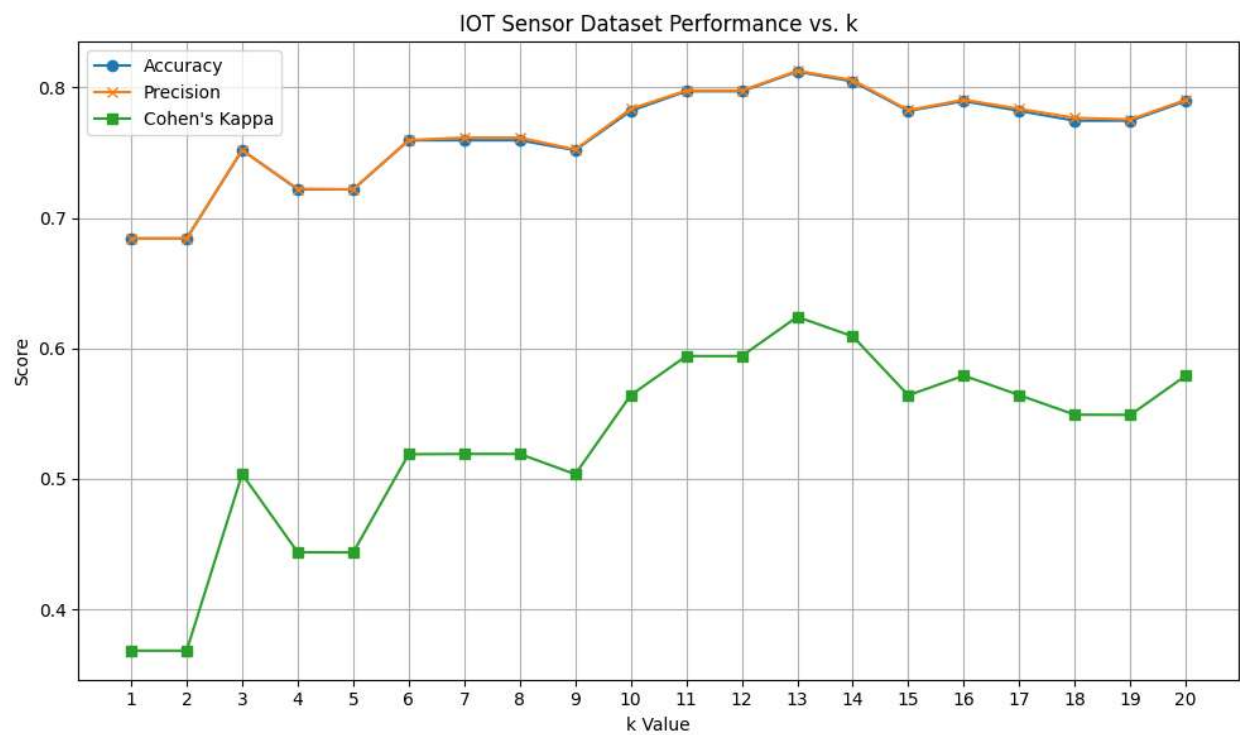
--- 2. Single Feature, Many Levels ---
Input: [{'Error_Code': 'E500'}]
Output Prediction: ['Critical']

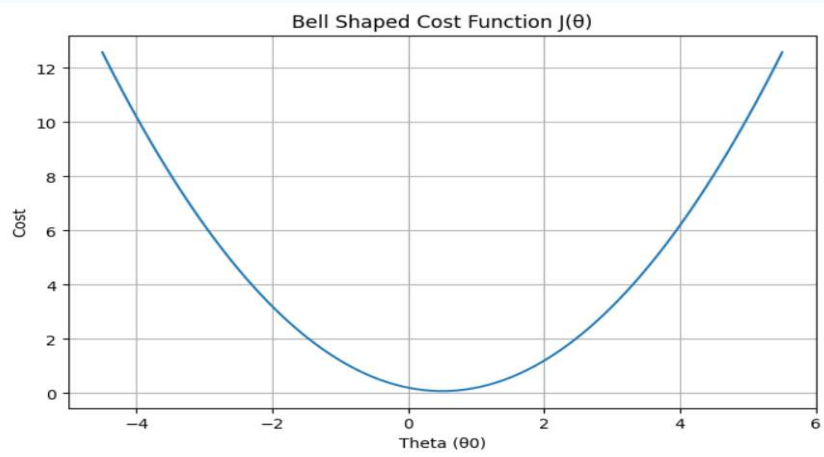
--- 3. Multiple Features, 2 Levels ---
Input: [{'Motion': 'Yes', 'Light': 'Off'}]
Output Prediction: ['No']

--- 4. Multiple Features, Many Levels ---
Input: [{'Protocol': 'HTTP', 'Location': 'Asia'}]
Output Prediction: ['High']
```



Optimal K value determined and classification performed.





Model Evaluation
Mean Squared Error: 0.1629
R2 Score: 0.3418